



United Nations
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International Center for Integrated
Water Resources Management
under the auspices of UNESCO

International Center for Integrated Water Resources Management (ICIWaRM): Update

For US National Committee for IHP
January 18, 2018, Washington, DC

Will Logan
Deputy Director



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International Center for Integrated Water Resources Management (ICIWaRM): Update

- G-WADI tools and plans
- Collaboration and Public Participation/Collaborative Modeling
 - SCORES
- CRIDA and Risk-Based Hydrologic Engineering Standards
 - Regions 1 and 3 Meetings
 - Communications

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African-Flood-and-Drought-Monitor finding Applications in Health, Epidemiology and Migration studies

Developed by Princeton University's [Terrestrial Hydrology Group](#) in collaboration with UNESCO IHP [G-WADI](#) and ICIWaRM, the [African Flood and Drought Monitor](#) (AFDM) monitors and forecasts meteorological, agricultural and hydrological drought at various temporal and spatial scales. It also has a multi-decadal, historical reconstruction of the terrestrial water cycle against which current conditions can be compared.

Some of the applications of the AFDM were foreseen by its developers, but researchers and international development specialists are finding ways to use the monitor in many creative ways. In the last two years alone, AFDM data and information have been used for:

Drought Resilience: Controlling for drought variables with the AFDM helped USAID establish that a) the overall shock exposure in Niger vs. Burkina Faso during their [RISE](#) study was due to insect invasions and food price increases, not climate; and b) their market-based interventions in the [PRIME](#) program in Ethiopia were effective.

Impact of Irrigation Dams: Use of AFDM drought indices helped show that in northern Nigeria households downstream of the dams were less affected by drought and enjoyed more stable growth rates and food consumption than those upstream or in similar but undammed basins.

Human migration: Rainfall shortages and excess temperature, with soil moisture as an additional important factor, are strong drivers of out-migration from South Africa, especially for black and low-income migrants.

Health and Epidemiology: a) Temperature drives both tsetse fly relative abundance and trypanosome prevalence in Tanzania; b) Risk of cholera transmission in Cameroon varies with average daily maximum temperature and with the precipitation levels over the preceding two weeks.



Stream.princeton.edu

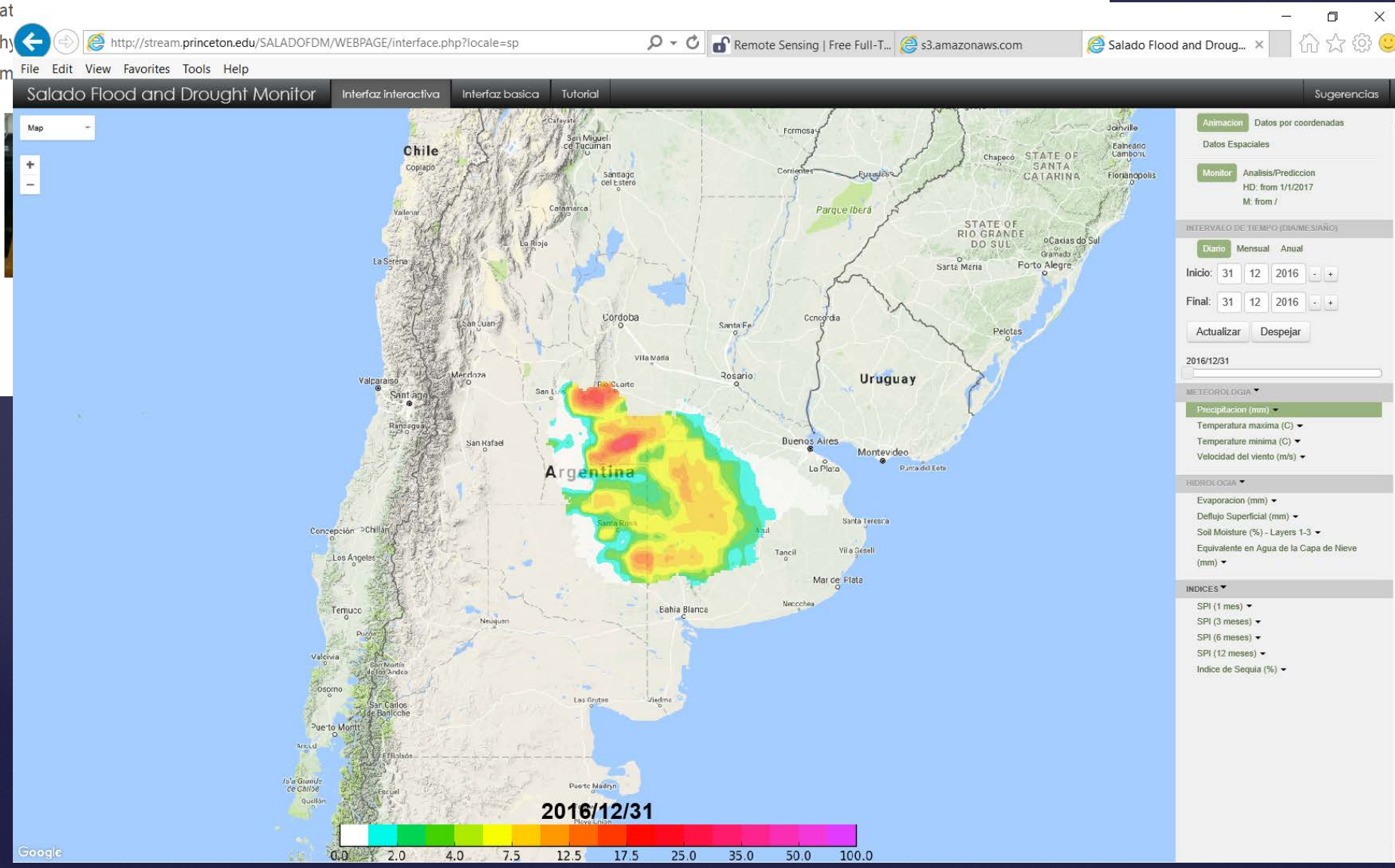
Advancing the Implementation of the Latin American Flood and Drought Monitor at the National Level

The [Latin American Flood and Drought Monitor](#), created by Princeton University's [Terrestrial Hydrology Research Group](#) with assistance by ICIWaRM and UNESCO-IHP's [G-WADI](#) program, is a regional product based on advanced land surface modeling driven by satellite and

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Improve Satellite Precipitation Estimation Using Gauge Observation

Bias adjustment of PERSIANN-CCS Estimation Using Local Gauge Measurements: A Case Study over Chile

❖ Gauge Observation

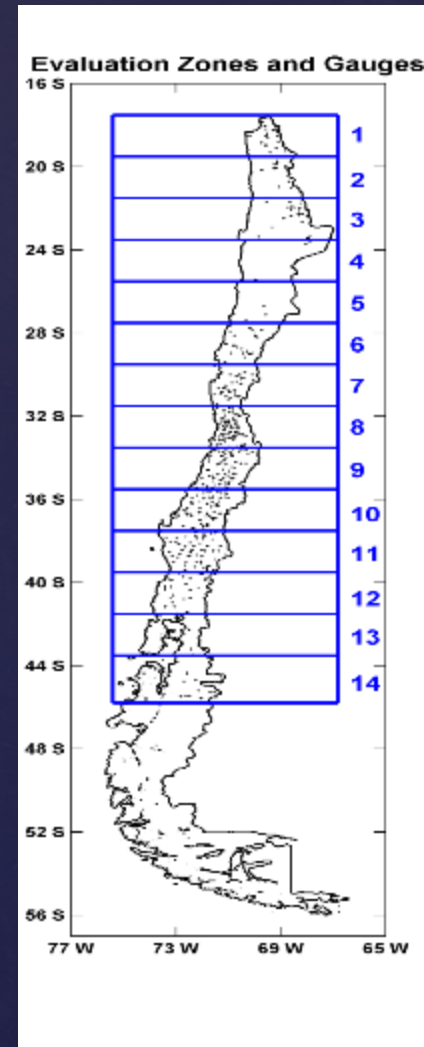
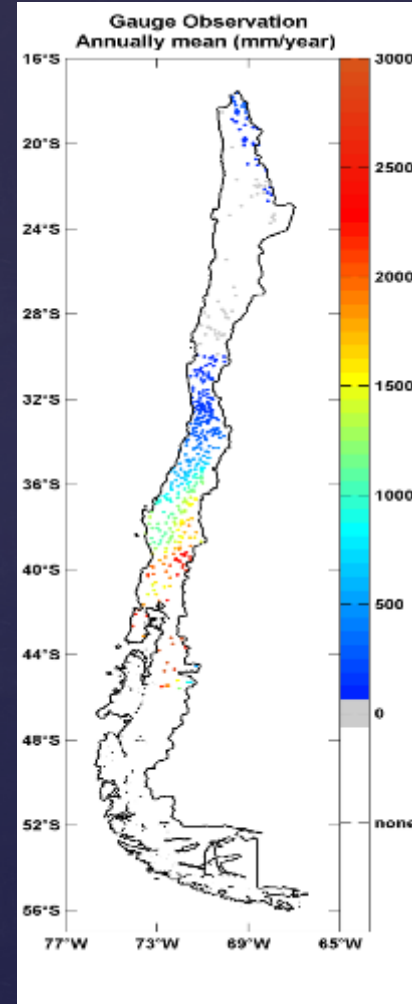
- 456 gauges are selected
- Data period: 2009-2013 (5 years)
- Observations (see figures on the right)

❖ PERSIANN-CCS

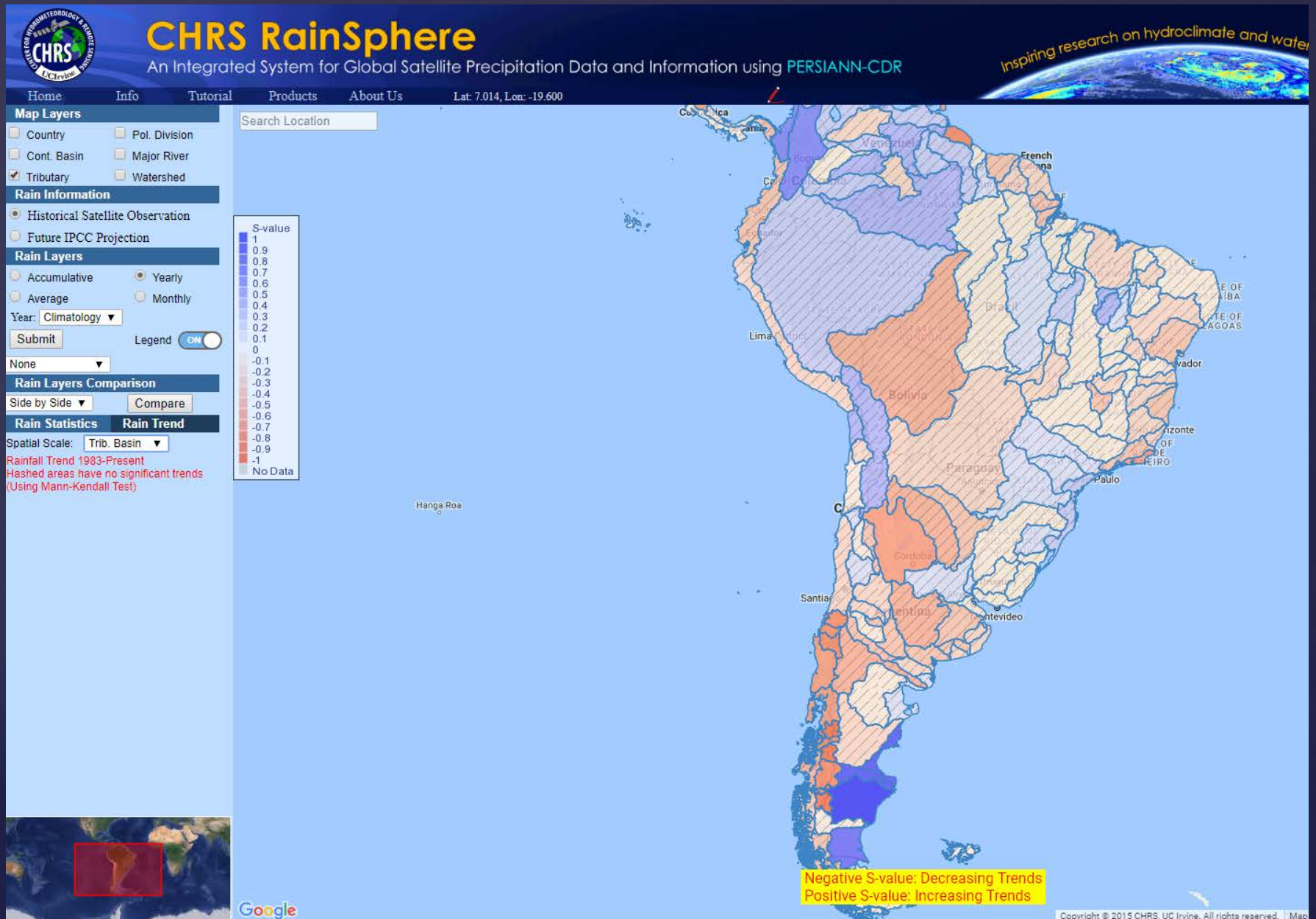
- Match to the same time period for evaluation

❖ Evaluation zones

- 14 zones are divided by every 2° latitudes over the adjusted region (shown as the right figure)



Precipitation Trends since 1983, by Tributary Basin

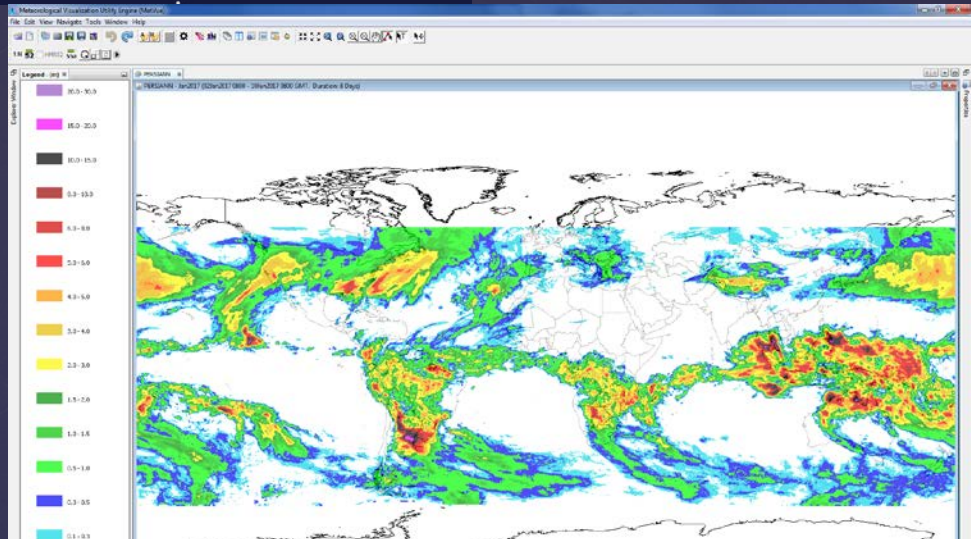




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Corps of Engineers' Hydrologic Engineering Center can now Import PERSIANN or other Gridded Rainfall Data into HEC-HMS (Hydrologic Model)



Google

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Dataset Time Step Domain

Visualization

Select DateTime to Format Compression

Able to download daily (e.g. 01Jan2017 – 10Jan2017) ArcGrid (*.asc) files for Whole Globe



Hydrologic Engineering Center

US Army Corps of Engineers

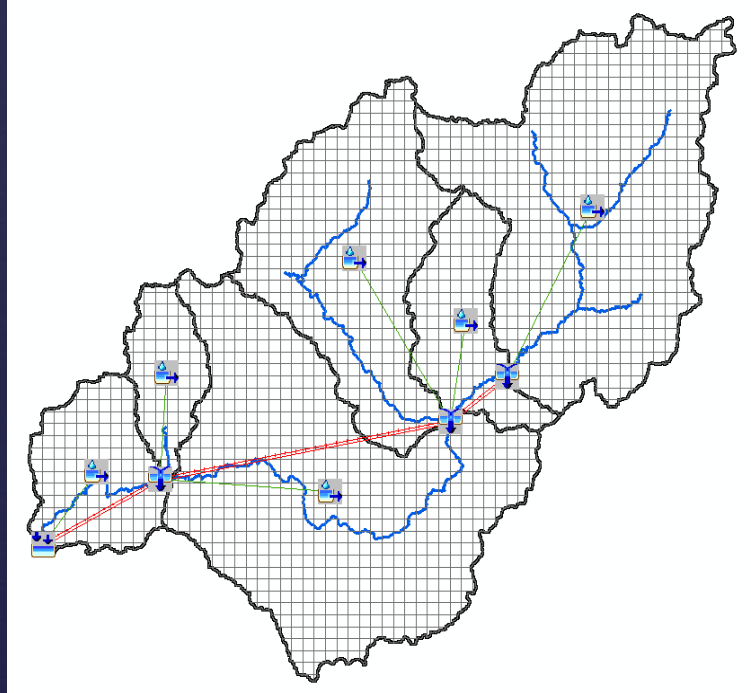
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Training

Training

Workshop: Application of HEC-HMS using Gridded Precipitation in Watersheds Outside of the United States (May 2017) This workshop is designed to help users applying HEC-GeoHMS and HEC-HMS in watersheds outside of the United States. In particular, this workshop will demonstrate the application of gridded precipitation within HEC-HMS outside of the regions covered by the HRAP and SHG grid systems. Prior to HEC-HMS version 4.2, it was not possible to utilize the gridded precipitation option in HEC-HMS for areas outside of the U.S. due to limitations in how gridded data was stored in HEC-DSS.

- [Download Workshop Document \(4.17 MB\)](#)
- [Download Program Setup Document \(164 KB\)](#)
- [Download ASCII Grid Import Export \(12.0 KB\)](#)
- [Download GageInterp User's Manual \(1.46 MB\)](#)
- [Download Workshop Files \(666 MB\)](#)



ICIWaRM Leads Collaborative Model Verification and Development Workshop for the Mekong River Commission



Workshop participants. Credit: CEO Pham Tuan Phan.

On 13-14 November 2017 in Vientiane, Lao PDR, ICIWaRM held a Collaborative Model Verification and Development Workshop for staff from the Mekong River Commission Secretariat (MRCS) and technical staff from the four National Mekong Committees (NMCs). The workshop was led by Drs. Seth Cohen and Dr. Jenny Olszewski.

ICIWaRM Subject Matter Experts Lead Shared Vision Planning Pilot in Tisza River Basin

At the request of the US Embassy in Hungary, a team of ICIWaRM subject matter experts is assisting in a Shared Vision Planning (SVP) Pilot to support water management by the JOINTISZA member nations (Hungary, Serbia, Slovakia, and Ukraine). These nations share the waters of the Tisza River Basin which is alternatively subject to severe urban flood risk and drought imperiling critical economic sectors. The uneven distribution and quality of water resources can magnify conflicts, generate population dislocations, and threaten fragile alliances in the region.

Shared Vision Planning is a tool and strategy which provides a structure for stakeholders to solve complex problems, and the basis for legitimizing the resulting solutions. Specifically, the SVP Pilot will help to balance the outcome of prior infrastructure investments in flood risk reduction by individual nations with projected future requirements for withdrawals across the river basin.



Tisza River Basin. Source: International Commission for the Protection of the Danube River.

INTERAGENCY AGREEMENT

Between the

U.S. DEPARTMENT OF STATE

(Bureau of Oceans and International Environmental and Scientific Affairs)

and the

U.S. Army Corps of Engineers (USACE), USACE's International Center for Integrated Water Resources Management (ICIWaRM)

For

Strengthening water COoperation for REgional Security (SCORES)

Goals and Objectives

Reduce conflict and promote regional stability and security by strengthening cooperation on shared waters in regions where water is, or may become, a source of conflict.

Foster economic growth and sustainable management of natural resources by

- Demonstrating best practices for IWRM
- Sound management of physical infrastructure and natural systems, and
- Improving basin-level governance of water resources.

ICIWaRM Conducts Collaborative Risk-Informed Decision Analysis (CRIDA) Technical Exchange at Asian Institute of Technology



Workshop participants conducting a "vulnerability assessment" exercise to define future conditions that could potentially "break" the system under consideration. Photo credit: Guillermo Mendoza.

Leaders from U.S.
Development P
lead by an ICIW
Thailand.

Second Workshop on "Water Resource Infrastructure Design in a Changing Environment" Dec 15-16, Washington DC

Arrivals: Thurs Dec 14 A Block of rooms, at Government rates, has been reserved for those coming from out-of-town, at the **Fairfax Hotel** [across the street from the Cosmos Club] under the name of '**Risk-Based Engineering Workshop**'

Phone: (202) 293-2100

Meeting Venue: Cosmos Club, 2121 Massachusetts Ave NW, Washington, DC 20008
<https://www.cosmosclub.org/splash>



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IHP Group 1 meeting: Perugia and Paris June 15, 16 and 19

- Timed to be right before IHP Bureau meeting
- Focused on preventing internal UNESCO reorganization that would demote IHP
- This turned out to be successful
- Canada and France have been reorganizing their national committees
- Growing number of Category 2 centers
- Growing number of UN-spawned water panels

XII Meeting of National Committees and Focal Points of the UNESCO International Hydrological Program for Latin America and the Caribbean (IHP-LAC)

(Nassau, The Bahamas, 18-20 September, 2017)

- Timed to dodge hurricanes...
- Category 2 centers and chairs played increased role, but Colombian C2C is formally deceased
- Worked a lot with the Chileans on various projects to apply G-WADI tools at a country- or sub-basin scale
- International Sediment Initiative (ISI), International Flood Initiative (IFI) and Ecohydrology program seem to be going well
- Mexican C2C at UNAM has been formally approved by General Conference. UNESCO chair in Puebla is active and healthy. Talk about collaboration with US continues.
- Next meeting in Panama, October 2018 (sic)



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ICIWaRM and G-WADI website migration

- USA.gov sites are being discontinued, so have to migrate ICIWaRM
- University of Arizona has migrated GWADI.org to a new server. Completed except for tweaks.